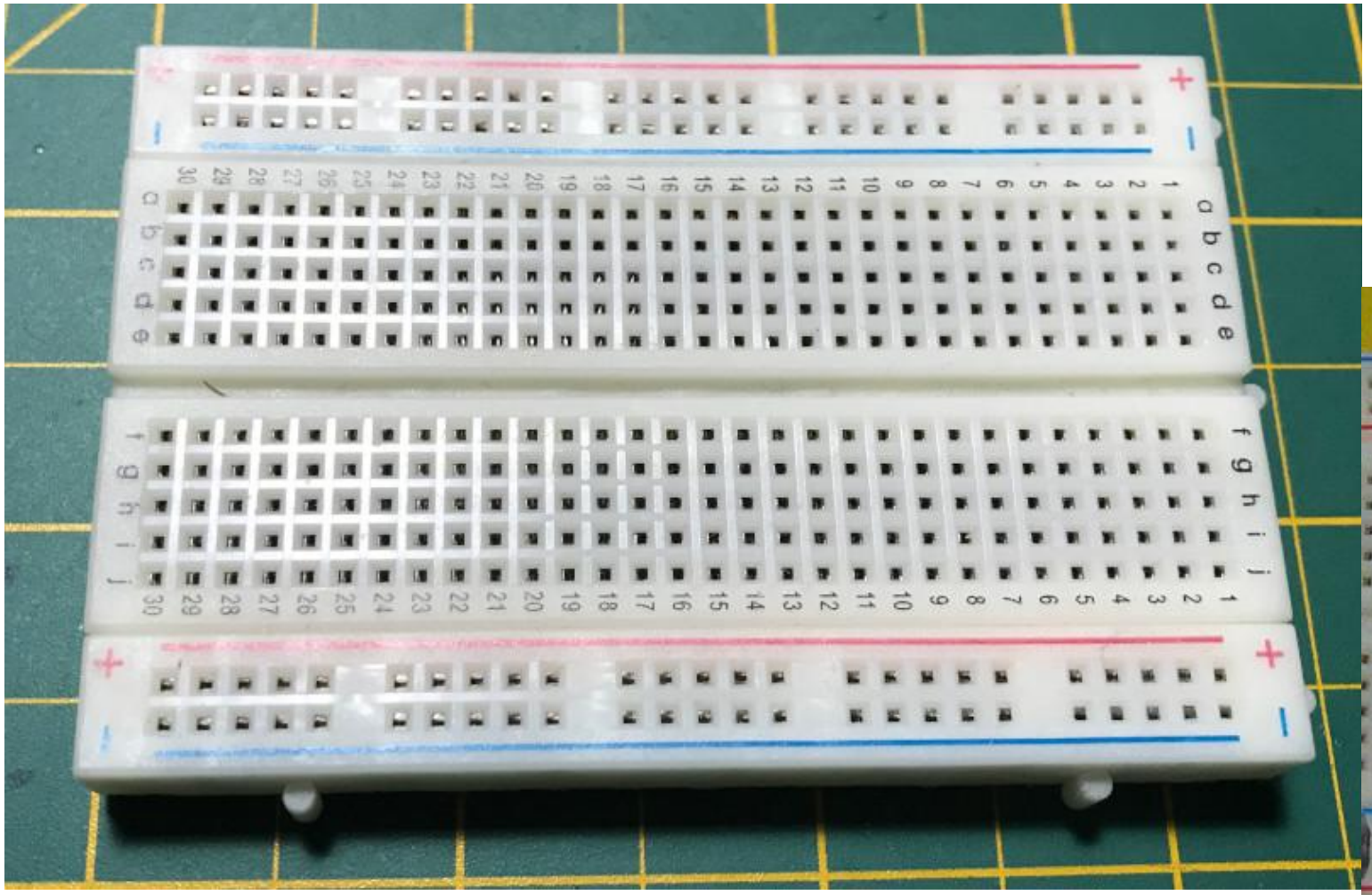
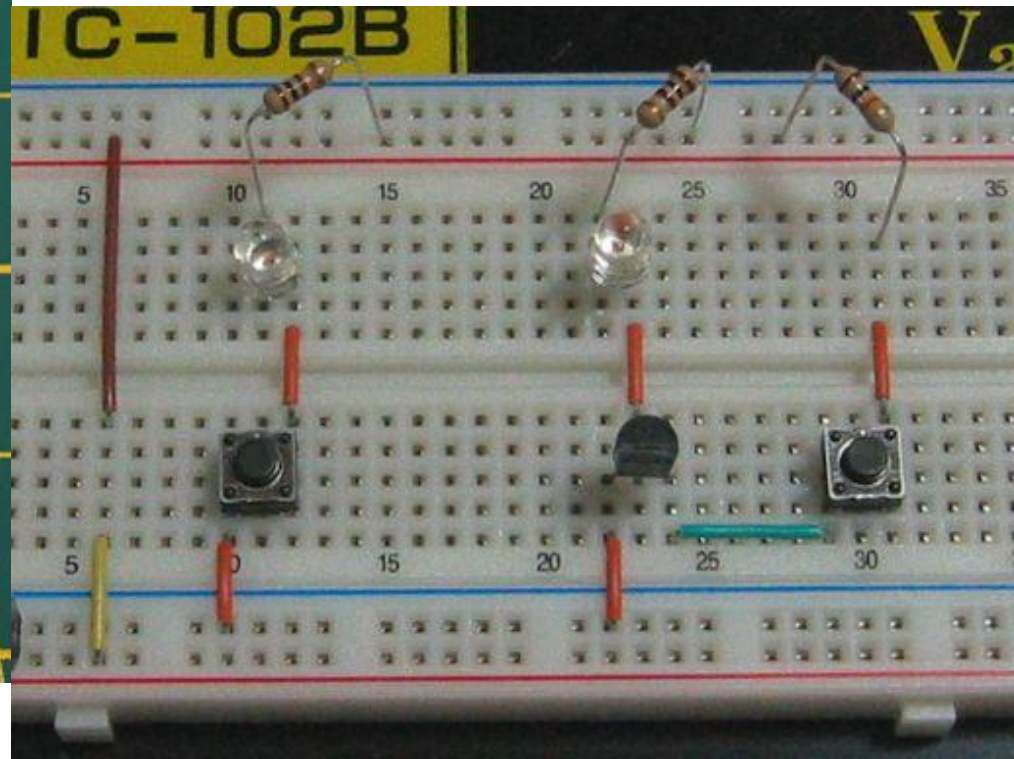
第３回講習資料

1.1

- 1. ブレッドボード上でタイマーicを使ってみる

1.1 ブレッドボード

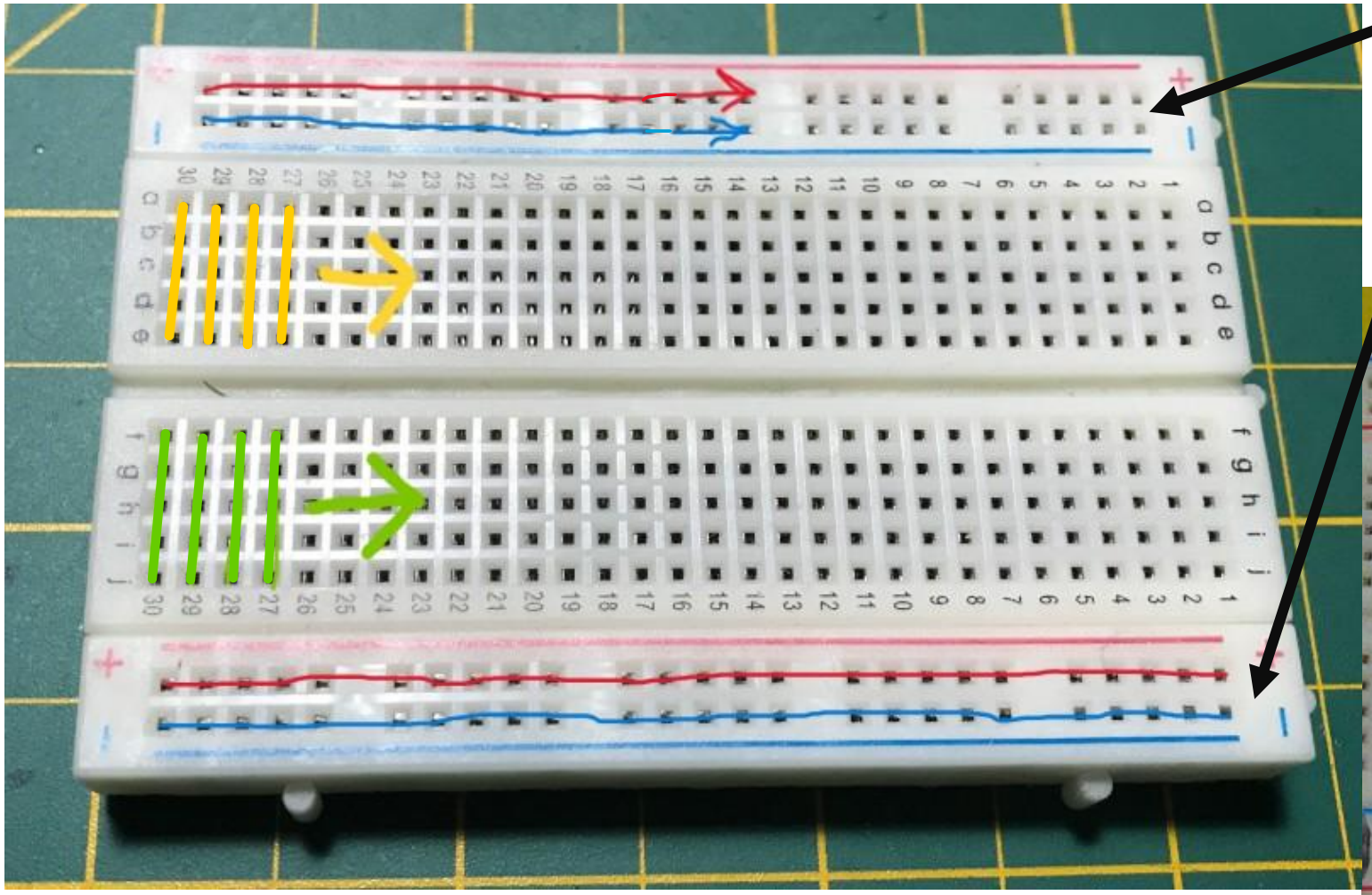


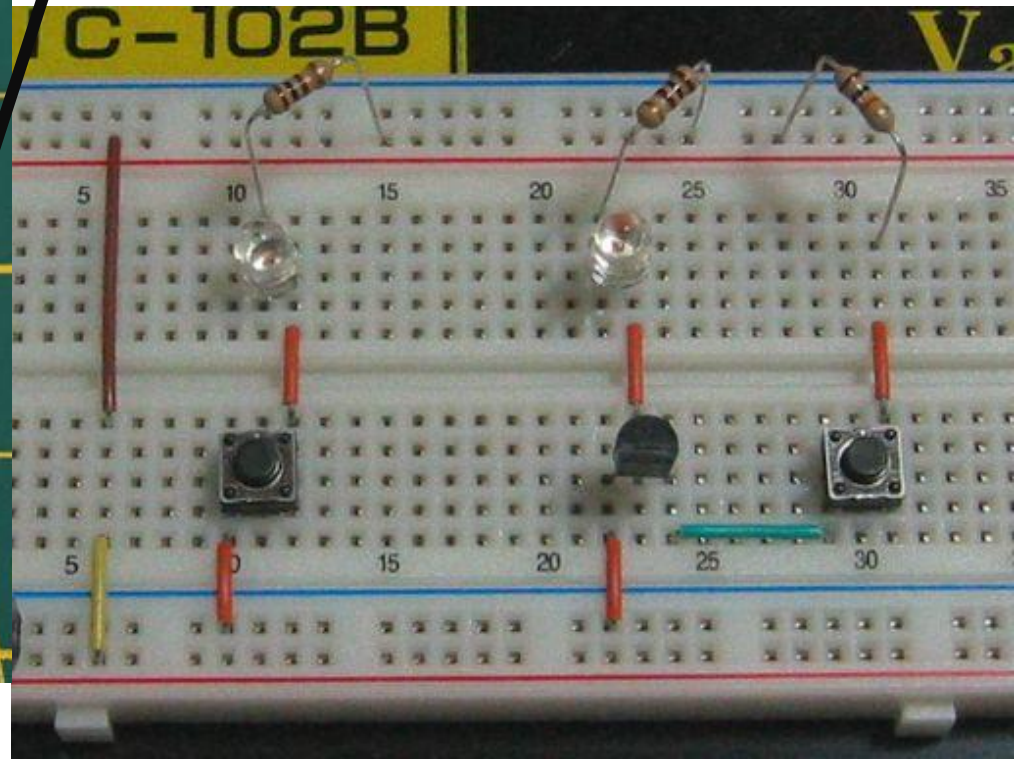


参照元 https://iot.keicode.com/electronics/what-is-breadboard.php

1.1 ブレッドボード

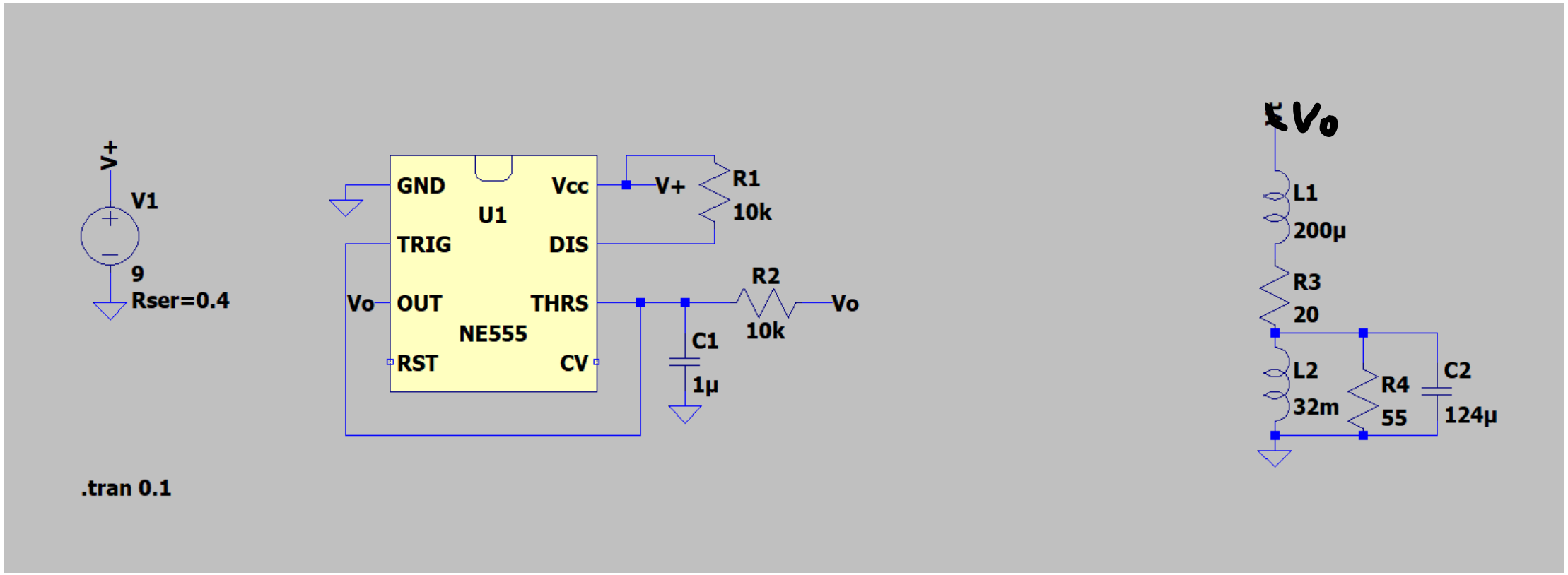
主に電源の配線に使う





参照元 https://iot.keicode.com/electronics/what-is-breadboard.php

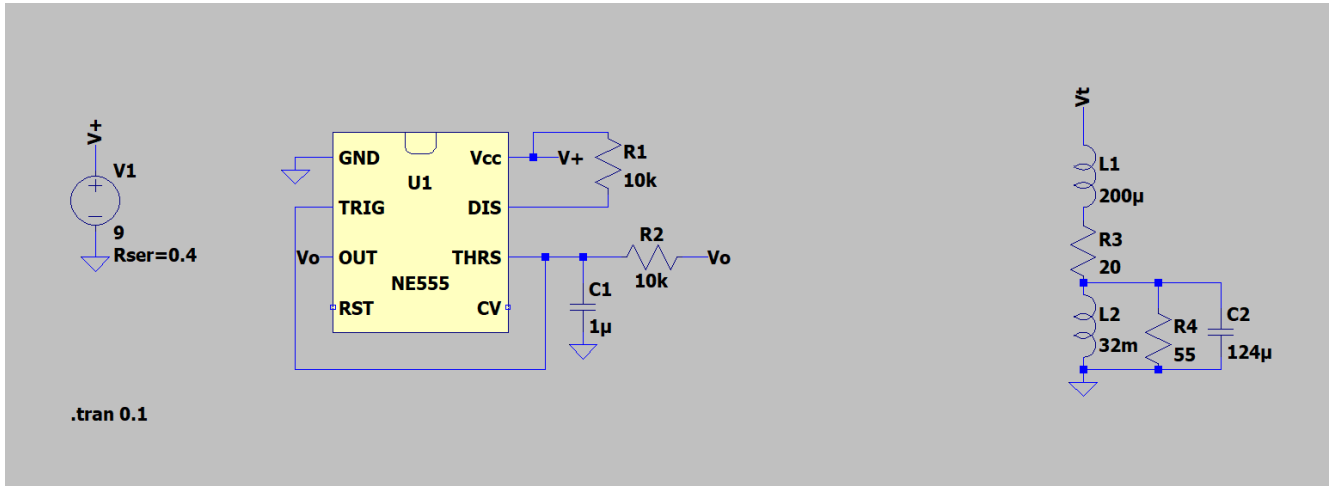
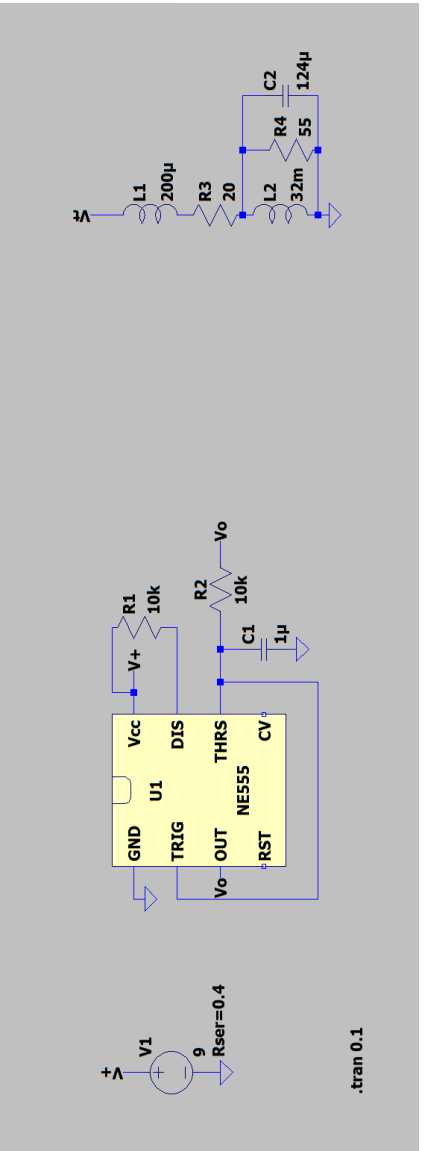
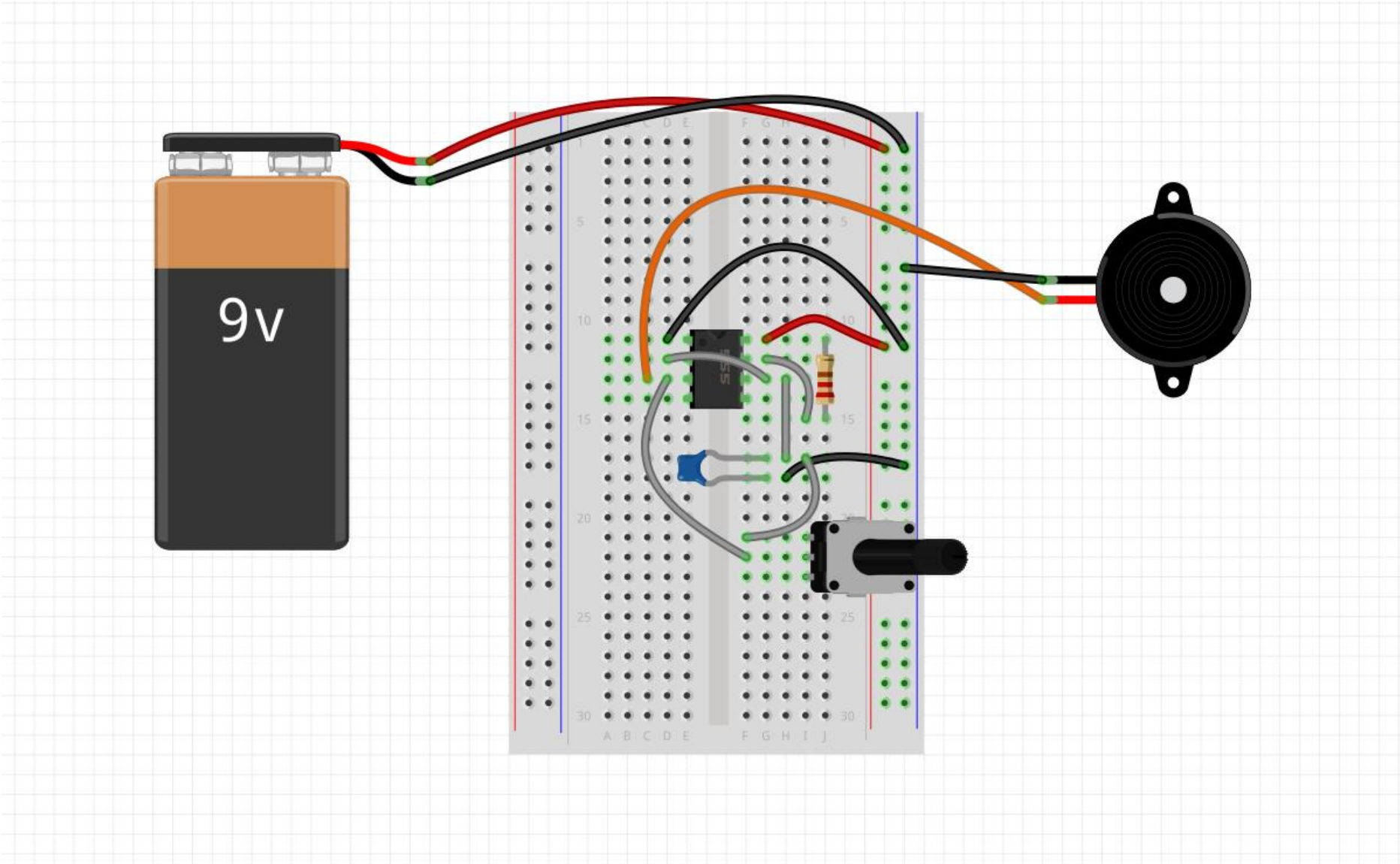
1.2 回路図


V+
V1
9
Rser=0.4
GND
Vcc
U1
TRIG
DIS
Vo
OUT
THRS
NE555
RST
CV
V+
R1
10k
R2
10k
Vo
C1
1μ
.tran 0.1
Vo
L1
200μ
R3
20
L2
32m
R4
55
C2
124μ

1.2 回路図

NE555：有名なタイマーIC
抵抗、コンデンサ(ダイオード等)の簡単な組み合わせで様々なパルス信号を生成できる
今回は周期的にパルスを発生させて音を作る

回路の右側：スピーカーの等価回路
スピーカーの中身は抵抗とコイル、コンデンサの組み合わせで表現できる



1.2 回路図をブレッドボードで表現